

ECE 3337: Final Exam

Spring 2026

Last name of student POSTED SOLUTIONS	Student ID number 
First name of student 	Email address 

Do This First:

1. Make sure that you have all the pages of the exam book.
2. Fill in the above information.
3. Turn off and put away all electronic gadgets (laptop, tablet, smartphone, calculator, smartwatch, wearable, ...)

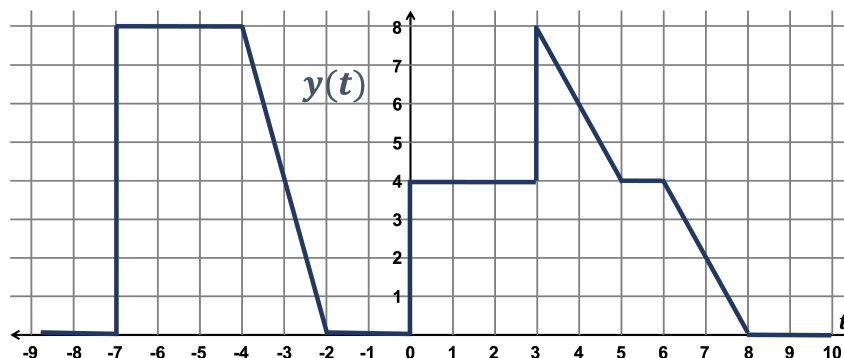
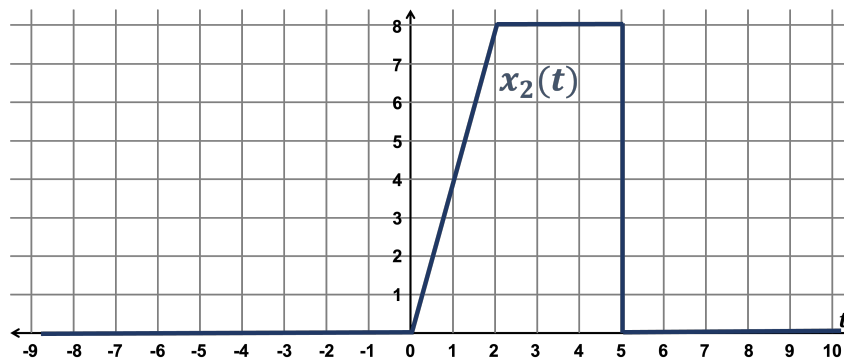
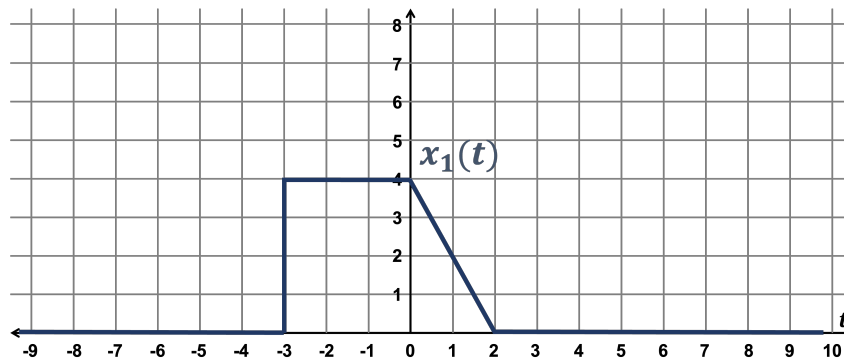
Exam Rules:

1. This is a 2-hour exam.
2. The syllabus includes all material covered in class.
3. You are allowed to bring two 8.5" x 11" handwritten crib sheets (double-sided).
4. If you need additional space to answer a question, continue on the back page(s).
5. You are not allowed to interact with any other student.

Q1	Q2	Q3	Q4	Q5	Q6	Total
15	10	10	20	25	20	100
						

1. (15 points) Waveforms and Convolution

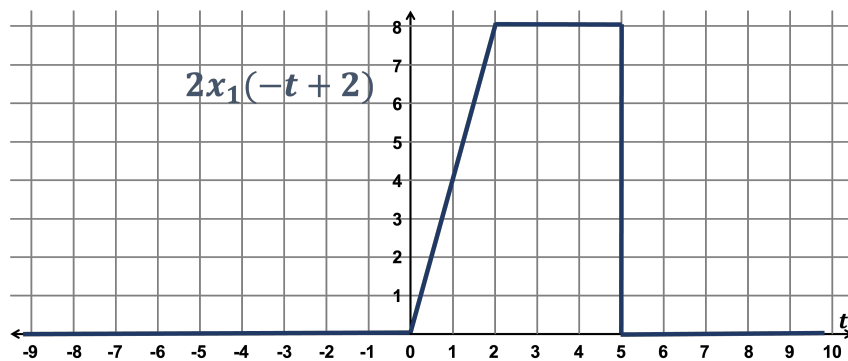
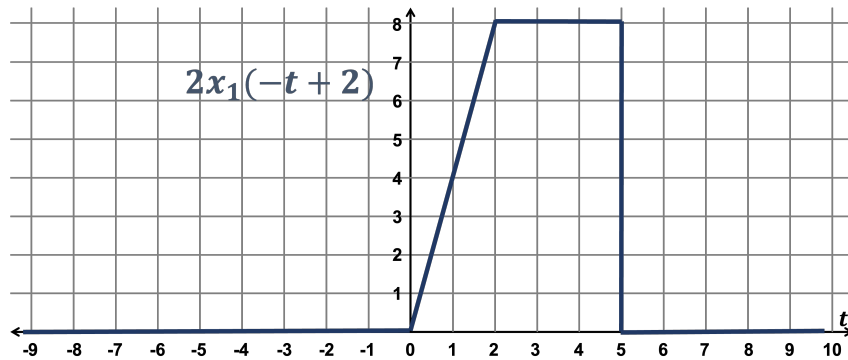
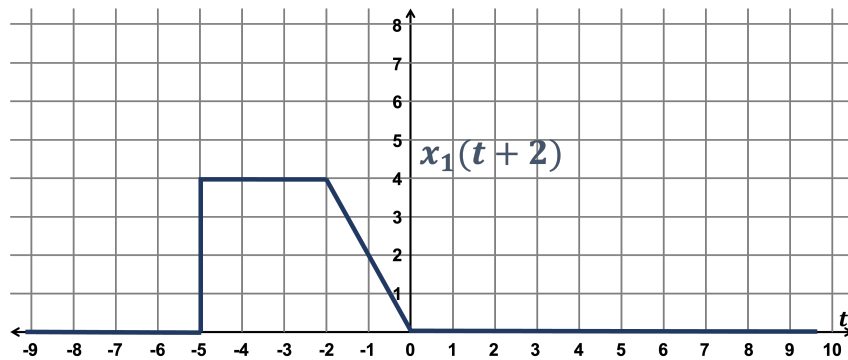
The signals $x_1(t)$, $x_2(t)$, and $y(t)$ are as graphed below.



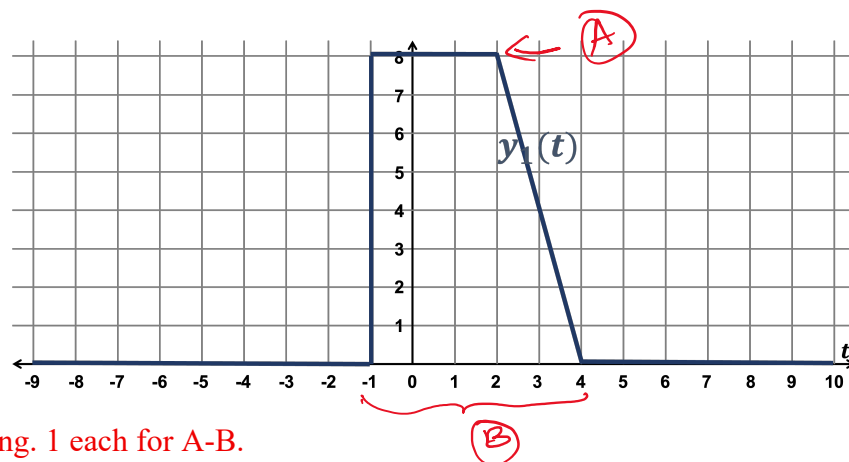
- a. (5 points) The signal $x_2(t)$ is a transformed version of $x_1(t)$ such that $x_2(t) = C x_1(At + B)$. Find the values of A , B and C .

$$\begin{aligned}
 A &= \underline{-1} \\
 B &= \underline{2} \\
 C &= \underline{2}
 \end{aligned}$$

1 point for trying. 1 each for A-D.

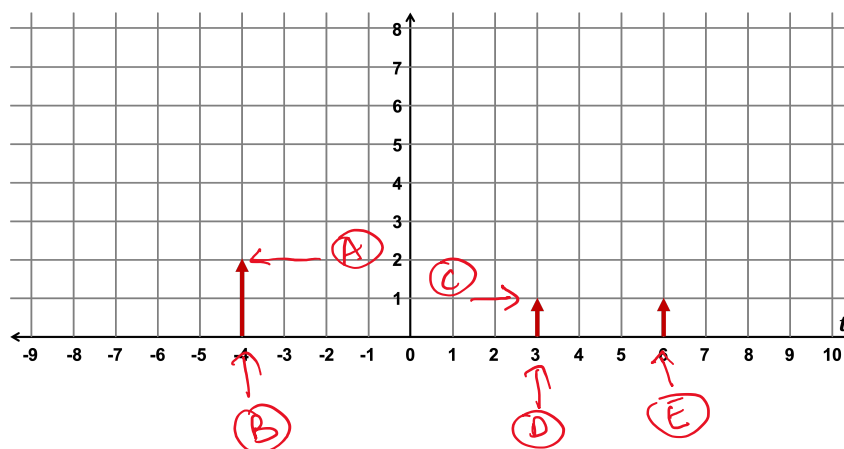


- b. (3 points) Graph the output of the convolution: $y_1(t) = x_1(t) * 2\delta(t-2)$, where $*$ represents convolution. Your final answer must be presented in the space below, and this will be the only answer graded.



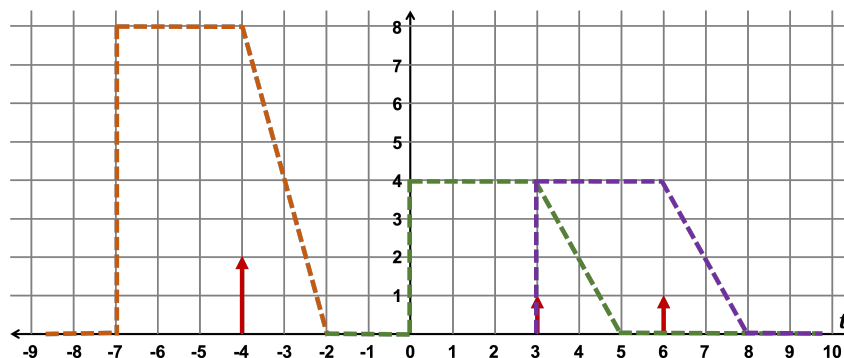
1 point for trying. 1 each for A-B.

- c. (7 points) If the signal $y(t)$ graphed above satisfies the equation: $y(t) = x_1(t) * z(t)$, where $*$ represents convolution, then graph the signal $z(t)$. Your final answer must be presented in the space below, and this will be the only answer graded.



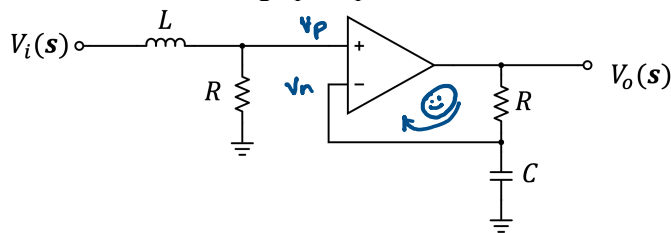
2 points for trying. 1 each for A-E.

Adding the following will give $y(t)$.



2. (10 points) Linear Time-invariant Systems

Consider the following op-amp circuit:



- a. (4 points) Derive a symbolic expression in terms of R, L, C for the magnitude of the transfer function $|H(\omega)|$

$$V_p = \frac{R}{R + j\omega L} V_i = V_n = \frac{j\omega C}{R + j\omega C} V_o = \frac{1}{(1 + j\omega CR)} V_o \Rightarrow \frac{V_o}{V_i} = H(\omega) = \frac{1 + j\omega CR}{1 + j\omega \frac{R}{L}}$$

$$\Rightarrow |H(\omega)| = \frac{\sqrt{1 + (\omega RC)^2}}{\sqrt{1 + \left(\frac{\omega}{R/L}\right)^2}}$$

Grading: -2 pts trying

+2 pts correct $|H(\omega)|$

- b. (3 points) Identify the poles and zeros for this system for $R = 1 \Omega, L = 0.1 H, C = 10 mF$.

pole at R/L , zero at $1/RC$

$$\frac{R}{L} = \frac{1}{0.1} = 10 \text{ rad/s}$$

(A)

$$\frac{1}{RC} = \frac{1}{10 \times 10^{-3}} = 100 \text{ rad/s}$$

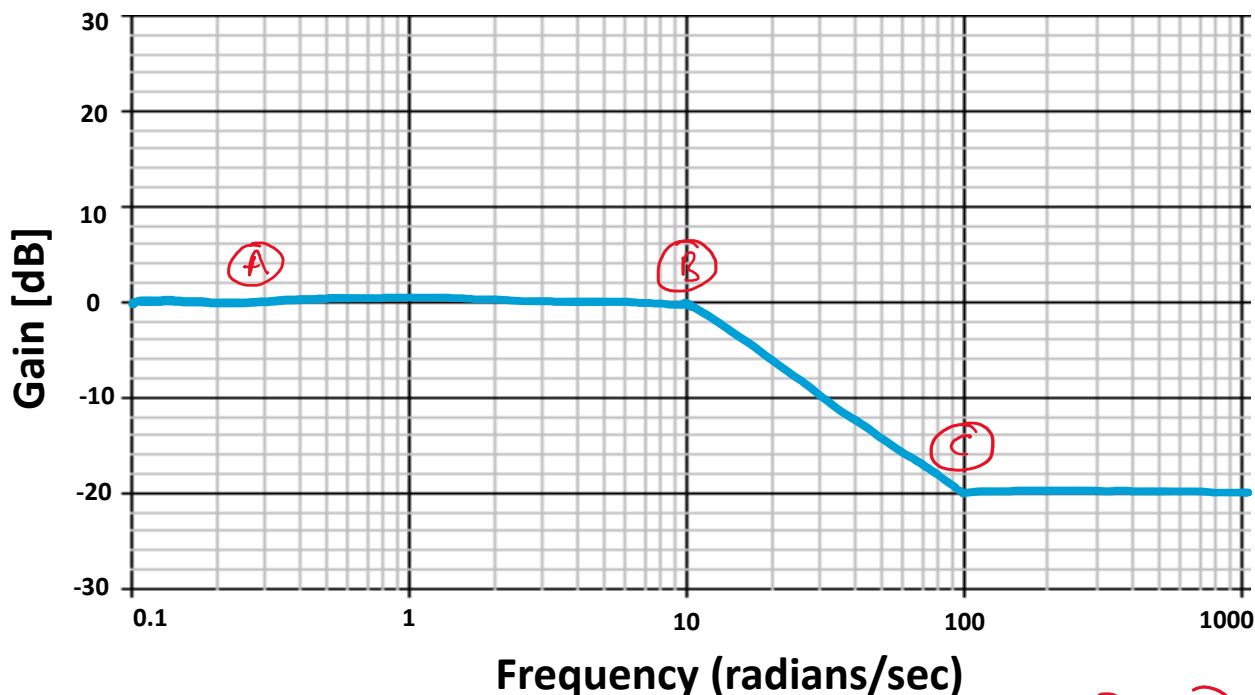
(B)

Grading:

• 1 pt trying

• +1 pt each (A), (B)

- c. (3 points) Sketch the Bode magnitude plot for this system below.



Grading: 1 pt each part (A), (B), (C)

3. (10 points) Laplace Transform & Its Inverse

a. (5 points) Determine the Laplace transform $\mathbf{X(s)}$ of the signal:

$$x(t) = e^{-t} u(-t + 2) u(t).$$

Note that $u(-t + 2) = 1$ when $t < 2$ and 0 for $t > 2$.

Similarly, $u(t) = 1$ when $t > 0$ and 0 for $t < 0$

Therefore, $u(-t + 2) u(t) = 1$ only when $0 < t < 2$, and 0 otherwise.

(A)

$x(t)$ can be rewritten as

$$\begin{aligned} x(t) &= e^{-t} [u(t) - u(t - 2)] = e^{-t} u(t) - e^{-t} u(t - 2) \\ &= e^{-t} u(t) - e^{-2} e^{-(t-2)} u(t - 2) \end{aligned} \quad - \quad (B)$$

Taking the Laplace transform,

$$\mathbf{X(s)} = \frac{1}{s+1} - \frac{1}{s+1} e^{-2(s+1)}$$

(C) (D)

1 point for trying. 1 each for A-D.

b. (5 points) Determine the inverse Laplace Transform of $\mathbf{H(s)} = \frac{s^3 - 4s}{s^2 + 8s + 20}$.

Performing long division,

$$\mathbf{H(s)} = \frac{s^3 - 4s}{s^2 + 8s + 20} = s - 8 + 40 \frac{s + 4}{s^2 + 8s + 20} \quad - \quad (A)$$

Note that $s^2 + 8s + 20 = (s + 4)^2 + 2^2$ - (B)

Therefore,

$$\mathbf{H(s)} = s - 8 + 40 \frac{(s + 4)}{(s + 4)^2 + 2^2}$$

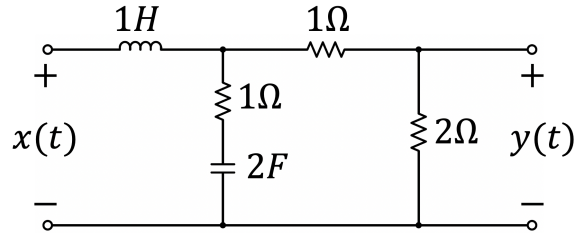
Taking the inverse Laplace of each term

$$h(t) = \frac{\delta'(t) - 8\delta(t)}{(C)} + \frac{40e^{-4t} \cos(2t) u(t)}{(D)}$$

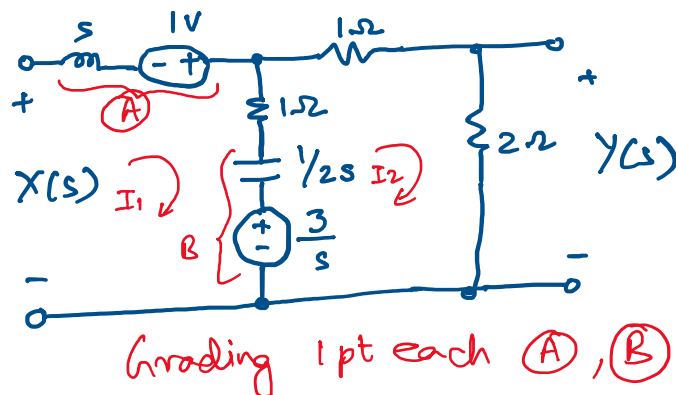
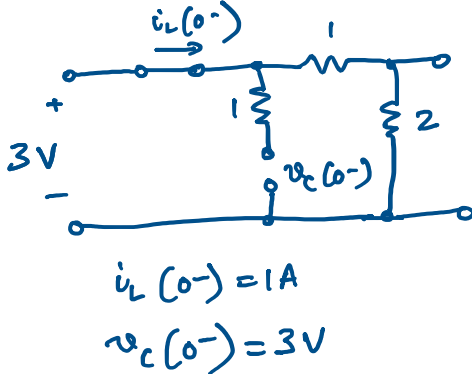
1 point for trying. 1 each for A-D.

4. (20 points) Applications of the Laplace Transform

Consider the following circuit and the input signal $x(t) = 3u(-t) + \sin(t)u(t)$. Use the back sheet for your work as needed and write the final answers in the spaces below.



- a. (2 points) Sketch the Laplace-domain version of the given circuit with the initial conditions indicated.



- b. (1 point) Write the Laplace transform of the input $X(s)$ for $t \geq 0$.

$$X(s) = \frac{1}{s^2 + 1}$$

Grading 1 pt correct answer

- c. (7 points) Calculate the output $Y(s)$ for a generic input $X(s)$ for $t \geq 0$. Your answer should be written as a ratio of polynomials in s .

(A)
$$\begin{cases} X(s) + 1 - \frac{3}{s} = I_1 \left(s + 1 + \frac{1}{2s} \right) - I_2 \left(1 + \frac{1}{2s} \right) \\ \frac{3}{s} = -I_1 \left(1 + \frac{1}{2s} \right) + I_2 \left(\frac{1}{2s} + 4 \right) \end{cases}$$

Grading: - 1 pt trying

- + 3 pt (A)
- + 2 pt (B)
- + 1 pt (C)

(B)
$$\Delta = A_1 A_4 - A_3 A_2 = \frac{8s^2 + 7s + 3}{2s}$$

(C)
$$\Delta_2 = A_1 V_2 - A_3 V_1 = \frac{8s + 1 + X(2s + 1)}{2s}$$

$$\Rightarrow I_2 = \frac{\Delta_2}{\Delta} = \frac{8s + 1 + X(2s + 1)}{8s^2 + 7s + 3}$$

$$\Rightarrow Y_2 = 2I_2 = 2 \left(\frac{8s + 1 + X(2s + 1)}{8s^2 + 7s + 3} \right)$$

- d. An LTI system has the following differential equation description, with initial conditions $x(0^-) = 1$, $y(0^-) = 1$, $x'(0^-) = 1$, $y'(0^-) = 1$, $y''(0^-) = 1$, and input $x(t) = u(t)$:

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 20y = \frac{dx}{dt} + x.$$

$$\hookrightarrow X(s) = \frac{1}{s}$$

- (i) (3 points) Identify the poles, zeros, and modes of this system.

$$s^2 + 12s + 20 = (s+2)(s+10) \Rightarrow \text{poles @ } -2, -10 \quad \textcircled{A}$$

$$\text{zero @ } -1 \quad \textcircled{B}$$

$$\text{modes @ } -2, -10 \quad \textcircled{C}$$

Grading: 1 pt each $\textcircled{A}, \textcircled{B}, \textcircled{C}$

- (ii) (7 points) Calculate the steady-state response of this system. Use the back sheet for your work, if needed, and write the final answer below.

convert the diff. equation to Laplace domain:

$$\textcircled{A} (s^2Y - s - 1) + 12(sY - 1) + 20Y = (sX - 1) + X = (s+1)X - 1$$

$$(s^2 + 12s + 20)Y = (s+1)X - 1 + s + 1 + 12 = (s+1)X + 12 = \frac{s+1}{s} + 12 = \left(\frac{13s+1}{s}\right)$$

$$\Rightarrow Y(s) = \frac{(13s+1)}{s(s+2)(s+10)} \quad \textcircled{B}$$

$$\Rightarrow \text{steady-state value} = \lim_{s \rightarrow 0} \{sY(s)\} = \lim_{s \rightarrow 0} \frac{13s+1}{(s+2)(s+10)} = \frac{1}{20} \quad \textcircled{C}$$

steady-state response

$$y_{ss}(t) = \frac{1}{20}u(t) \quad \textcircled{C}$$

Grading: • 2 pts for trying

• 3 pts for B

• 2 pts for $\textcircled{C}$ in either domain

5. (25 points) Fourier Series and Transform

a. (8 points). The Fourier series of a signal $x(t)$ is

$$x(t) = \sum_{n=-\infty}^{\infty} \frac{1}{(n^2 + 1)} e^{-(3+jn4)} e^{jnt}$$

Find the Fourier series of $x(t)$ in cosine/sine form. Please write your final answer here:

$$a_0 = \underline{\quad e^{-3} \quad (n = 0) \quad \text{A} \quad}$$

$$a_n = \underline{\quad \frac{2e^{-3}}{(n^2+1)} \cos(n4) \quad \text{B} \quad}$$

$$b_n = \underline{\quad \frac{2e^{-3}}{(n^2+1)} \sin(n4) \quad \text{C} \quad}$$

The relation between the Fourier series in the exponential and cosine/sin form is

$$x_n = \frac{1}{2}(a_n - jb_n).$$

Here,

$$\begin{aligned} x_n &= \frac{1}{(n^2 + 1)} e^{-(3+jn4)} = \frac{e^{-3}}{(n^2 + 1)} e^{-jn4} \\ &= \frac{e^{-3}}{(n^2 + 1)} [\cos(n4) - j \sin(n4)] \end{aligned}$$

Comparing $x_n = \frac{1}{2}(a_n - jb_n)$ and the above equation,

$$\begin{aligned} a_n &= \frac{2e^{-3}}{(n^2 + 1)} \cos(n4) \\ b_n &= \frac{2e^{-3}}{(n^2 + 1)} \sin(n4) \end{aligned}$$

For a_0 , substitute $n = 0$ in a_n .

2 points for trying. 2 each for A-C.

- b. (5 points) The Fourier transforms of time domain signals $x(t)$ and $y(t)$ are denoted by $X(\omega)$ and $Y(\omega)$, respectively. Find the Fourier transform of the signal $y(t) = x(t) \sin(t)$ in terms of $X(\omega)$.

Using Euler's identities,

$$y(t) = x(t) \sin(t) = \frac{1}{2j} x(t) [e^{jt} - e^{-jt}]$$

(A)

Using the Fourier frequency shifting property, (B)

$$x(t) e^{j\omega_0 t} \xleftrightarrow{\mathcal{F}} X(\omega - \omega_0)$$

Therefore,

$$x(t) e^{jt} \xleftrightarrow{\mathcal{F}} X(\omega - 1)$$

$$x(t) e^{-jt} \xleftrightarrow{\mathcal{F}} X(\omega + 1)$$

$$y(t) = \frac{1}{2j} x(t) [e^{jt} - e^{-jt}] \xleftrightarrow{\mathcal{F}} \frac{1}{2j} [X(\omega - 1) - X(\omega + 1)]$$

$$Y(\omega) = \frac{1}{2j} [X(\omega - 1) - X(\omega + 1)].$$

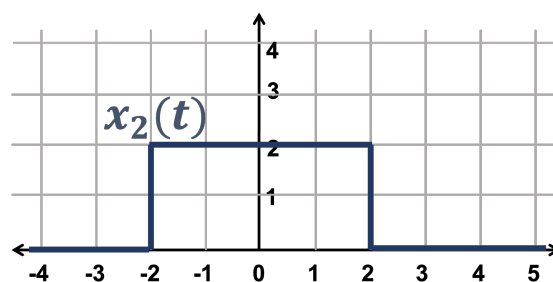
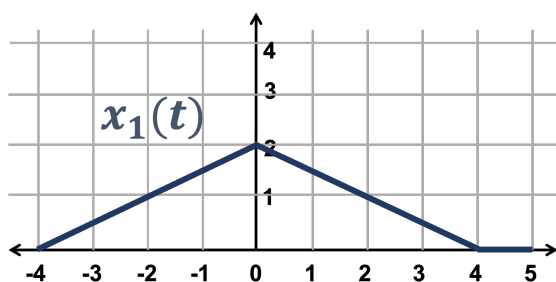
(C)

(D)

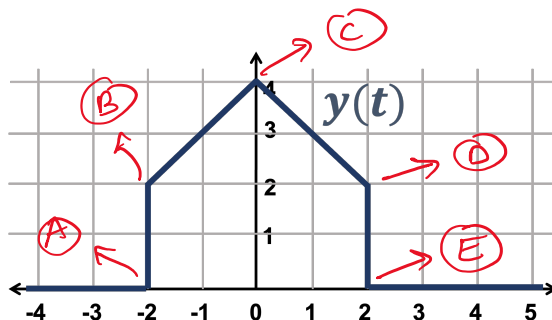
1 point for trying. 1 each for A-D.

- c. (6 points) The Fourier transforms of time domain signals $x_1(t)$, $x_2(t)$ and $y(t)$ are denoted by $X_1(\omega)$, $X_2(\omega)$, and $Y(\omega)$, respectively. Suppose $Y(\omega) = \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$, where $*$ represents convolution.

The signals $x_1(t)$, $x_2(t)$ are shown in the graphs below.



Sketch $y(t)$ below.



Using the convolution in frequency property of Fourier transforms,

$$x_1(t) \cdot x_2(t) \xleftrightarrow{\mathcal{F}} Y(\omega) = \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

In other words, multiplication in the time domain is equivalent to convolution in the frequency domain.

Therefore, we need to multiply the two functions to get $y(t)$.

$$y(t) = x_1(t) \cdot x_2(t)$$

The product is non-zero only in the range $-2 < t < 2$ and $x_1(t)$ is amplified 2 in this range.

1 point for trying. 1 each for A-E.

- d. (6 points) Find the Fourier transform of the signal $x(t) = u(t - 1) - u(t - 2)$.

Applying the definition of the Fourier transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

Since the signal is 1 in the range $1 < t < 2$ and zero otherwise,

$$\begin{aligned} X(\omega) &= \int_1^2 1 \cdot e^{-j\omega t} dt = \left. \frac{e^{-j\omega t}}{-j\omega} \right|_1^2 \\ &= \frac{j}{\omega} [e^{-j2\omega} - e^{-j\omega}] \end{aligned}$$

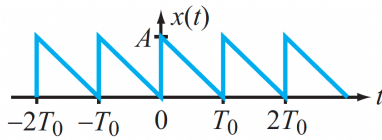
1 point for trying. 1 each for A-E.

This can also be simplified to

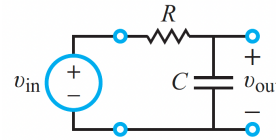
$$\begin{aligned} X(\omega) &= \frac{j}{\omega} e^{-\frac{j3\omega}{2}} [e^{\frac{j\omega}{2}} - e^{-\frac{j\omega}{2}}] \\ &= 2e^{-\frac{j3\omega}{2}} \left[\frac{e^{\frac{j\omega}{2}} - e^{-\frac{j\omega}{2}}}{2j\omega} \right] \\ &= e^{-\frac{j3\omega}{2}} \left[\frac{\sin\left(\frac{\omega}{2}\right)}{\left(\frac{\omega}{2}\right)} \right] \\ &= e^{-\frac{j3\omega}{2}} \text{sinc}\left(\frac{\omega}{2}\right) \end{aligned}$$

6. (20 points) Applications of Fourier Methods

- a. (5 points) Calculate the output $v_{out}(t)$ generated by the following RC circuit ($R = 1\Omega$, $C = 1F$) when the input signal is a backward sawtooth waveform for which the Fourier series is given below.



$$x(t) = \frac{A}{2} + \sum_{n=1}^{\infty} \frac{A}{n\pi} \sin\left(\frac{2n\pi t}{T_0}\right)$$



$$H(\omega) = \frac{1/j\omega C}{R + 1/j\omega C} = \frac{1}{1 + j\omega RC} \quad \text{(A)}$$

$$v_{out}(t) = \frac{A}{2} |H(\omega)| + \sum_{n=1}^{\infty} \frac{A}{n\pi} |H(n\omega_0)| \sin\left(\frac{2n\pi t}{T_0} - \tan^{-1}(\omega RC)\right)$$

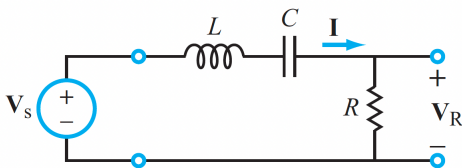
set $RC = 1$

$$= \frac{A}{2} + \sum_{n=1}^{\infty} \frac{A}{n\pi} \times \frac{1}{\sqrt{1 + (n\omega_0)^2}} \sin\left(n\omega_0 t - \underbrace{\tan^{-1}(n\omega_0)}_{\text{(D)}}\right)$$

Grading: (B) 1 pt for trying

+1 pt each (A), (B), (C), (D)

- b. (7 points) What type of filter does the following circuit implement? If $L = 10mH$ and $C = 1\mu F$, calculate the center frequency ω_0 , bandwidth B , and the value of R that will achieve a quality factor $Q = 10$.



$$\text{Bandpass filter } \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-2} \times 10^{-6}}} = 10^4 \text{ rad/sec} \quad \text{(A)}$$

Grading:

• 1 pt for trying

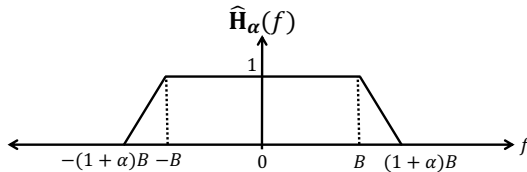
• +2 pts each (A), (B), (C)

$$Q = \frac{\omega_0}{B} \Rightarrow B = \frac{10^4}{10} = 10^3 \text{ rad/sec} \quad \text{(B)}$$

$$B = \frac{R}{L} \Rightarrow R = BL = 10^3 \times 10^{-2} = 10 \Omega \quad \text{(C)}$$

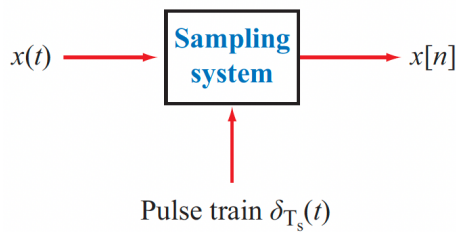
- c. (8 points) A marine biologist wishes to record the sounds produced by orcas, a species of whales, whose vocalization frequencies range up to 150kHz. Sketch the block diagram representation of a signal digitization system, along with the appropriate device settings that guarantee that the signal can be recovered without aliasing. The following devices are available.

- Low-pass filters with the following frequency response ($\alpha = 0.1$, B adjustable):

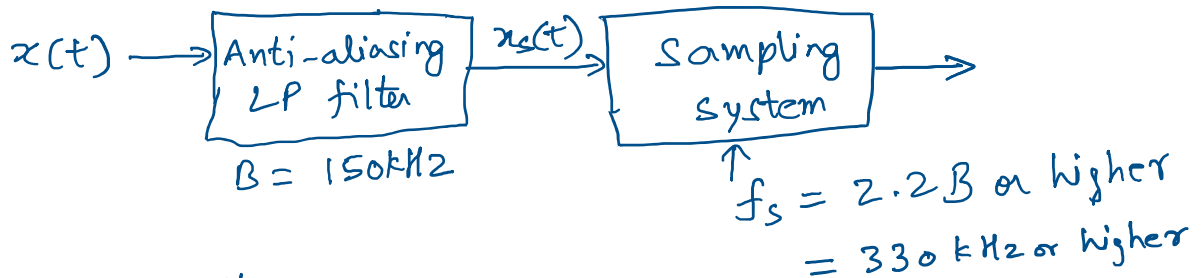


$$B = 150 \text{ kHz}$$

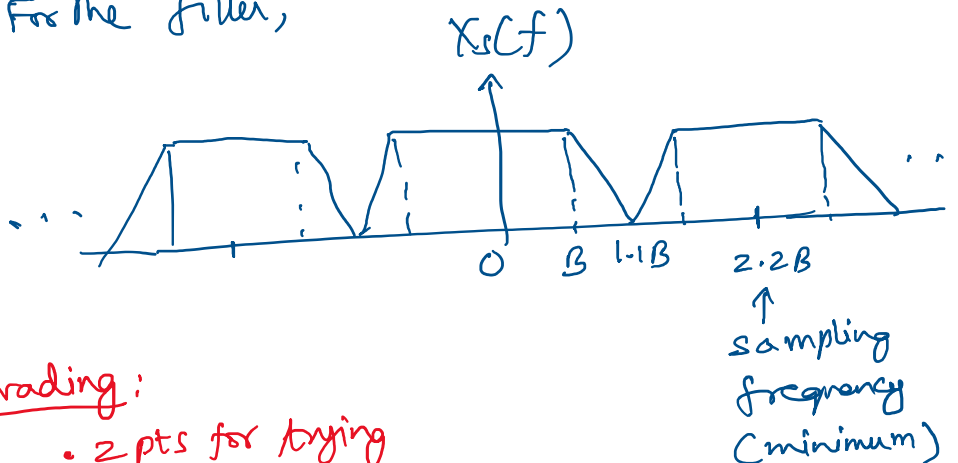
- A signal sampling system with an adjustable sampling rate of f_s Hz.



Block Diagram:



For the filter,



Grading:

- 2 pts for trying
- + 2 pts each correct B , f_s , & diagram